

# Xcelodose 600S- ME-5 Microbalance Failure Investigation

Mechanical failure pathway, observed handling practices, and corrective-action direction

RCA

AUG–SEP 2026



# The failure population separates into two distinct patterns

The mechanical pathway is concentrated in the units with visible deformation and corner-load failure.

4

microbalances reviewed

Unavailable between 2023–2026

2

have visible mechanical abnormalities and out-of-service

■ + ■

2

current direct-load observations

17-AUG + 20-AUG

1

most probable mechanical root cause

Handling control during cleaning

## Takeaway

Repeated direct force on the fragile weigh pan is the strongest supported mechanical pathway; alignment and powder conditions remain contributing pathways that should be controlled and monitored.

# Four failed / unavailable units

Mechanical Failure RCA basis upon historical evidence and observation.  
Software Failure will be alleviated in the next phase of the Replacement and Reliability Plan.

|                                                                                                                                                                         |                                                                                                                                                                                                      |                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><div></div><div>30-JUL-2026</div></div> <div>MECHANICAL</div> <div>Corner-load + linearity could not be adjusted</div> <div>Pan tilted left; multiple scuffs</div> | <div><div></div><div>24-AUG-2023</div></div> <div>MECHANICAL</div> <div>Failed corner load; could not be repaired</div> <div>TW#151494 05-SEP-2023</div> <div>Front springs visibly asymmetric</div> | <div><div></div><div>21-JUL-2026</div></div> <div>INTERFACE / ELECTRONIC</div> <div>Calibration software connection failure</div> <div>No visible tilt, scuffs, or external damage</div> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

RCA boundary    Mechanical root-cause analysis:  + ,  and  remain separate failure modes requiring their own follow-up.

# Mechanical abnormalities are visible on both corner-load failures

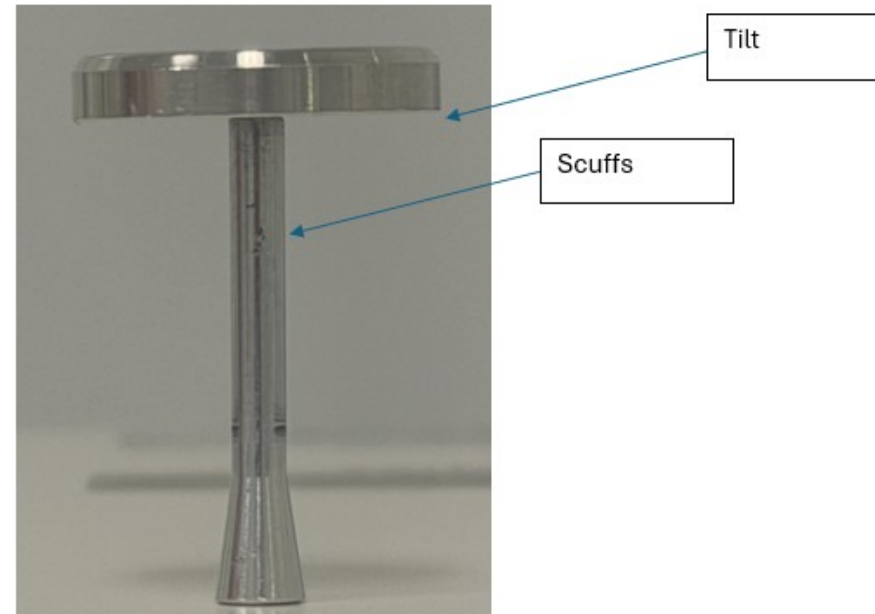
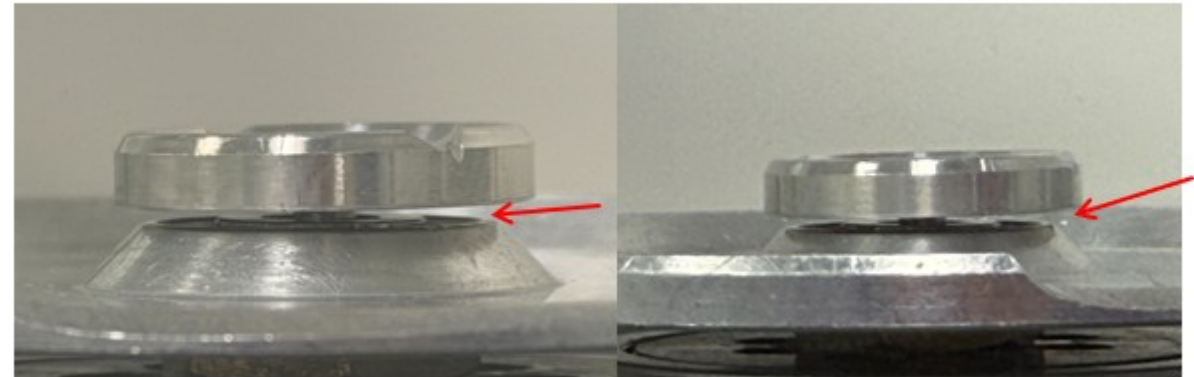
Pan tilted towards left on #0 [REDACTED]



## Tilted weigh pan + scuff marks

Failed manufacturer specification; corner load and linearity could not be adjusted.

- Visible geometry is consistent with deformation / misalignment.
- The exact time and loading event that produced the deformation is not directly observed.



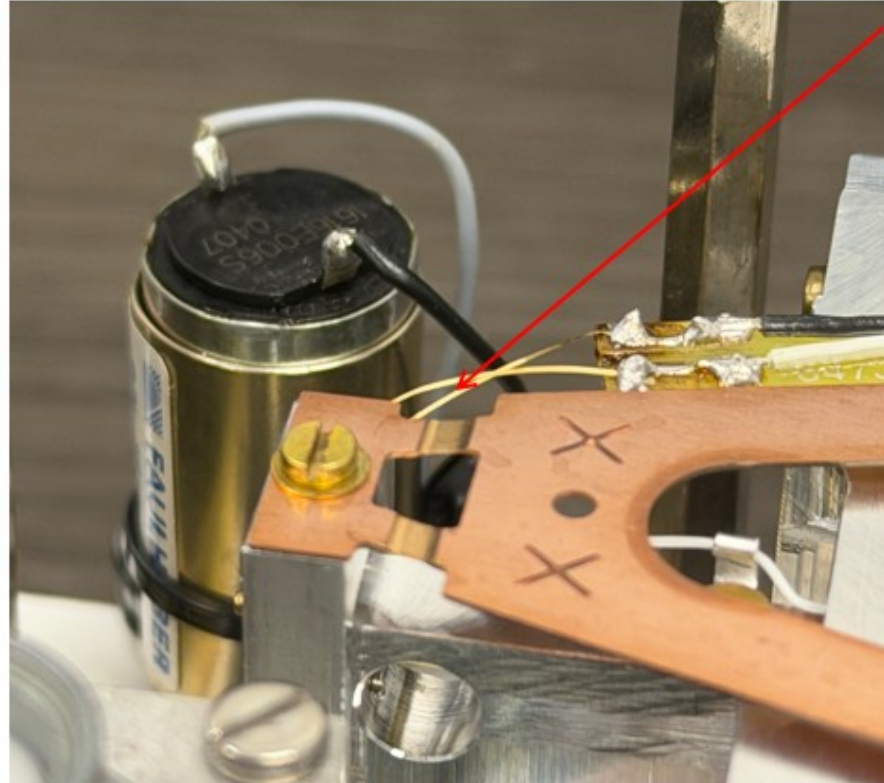
# Mechanical abnormalities are visible on both corner-load failures



## Asymmetric front springs

Failed corner-load test; unit could not be repaired.

- Visible asymmetry is consistent with spring deformation / mechanical misalignment.
- This unit provides historical evidence of the same corner-load failure family.
- This unit was investigated in TW#151494, direct loading was observed and determined as deformations cause; Pan bend observed similarly to # [REDACTED]



Top front springs are aligned differently, left slopes downward and right side slopes upward.

# Triangulate hardware evidence, observed practice, and controlled requirements

No single observation is treated as sufficient on its own, the mechanical cause was built by comparing multiple evidence streams.

## FAILED-UNIT INSPECTION

Tilt, scuffs, spring  
geometry

## MANUFACTURING GEMBA

Operation, cleaning,  
handling

## SOP / PRACTICE COMPARISON

What is required vs  
observed

## PRODUCT DEV COMPARISON

Different setup +  
cleaning practice

## HISTORY + SERVICE

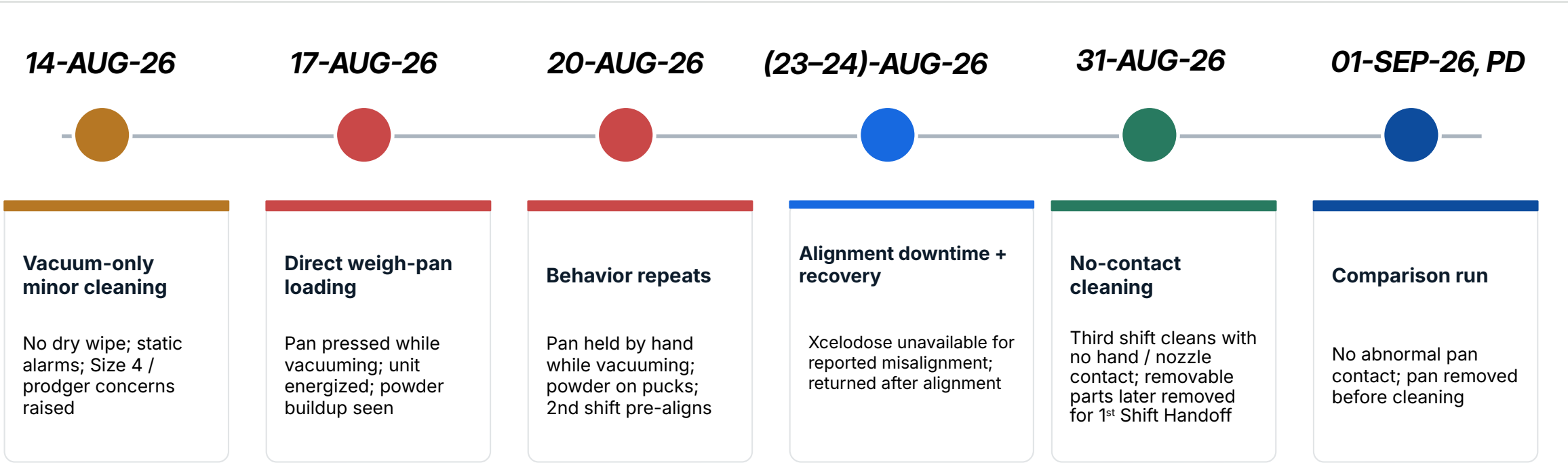
2023 event,  
Technician findings

## 6M + 5-WHY SYNTHESIS

Rank pathways +  
system cause

Evidence Flow | Historical Investigation → Observations and SOP Derivations → PD Observations and Comparisons

# The strongest evidence emerged across repeated cleaning and alignment observations



**Pattern across time** The critical handling condition was observed more than once, while later comparison runs demonstrated that cleaning can be performed without direct weigh-pan loading.

# Direct weigh-pan force creates a credible load path into the weighing mechanism

This is the only mechanical pathway supported by repeated direct observation plus a relevant historical event.

## DIRECT OBSERVATION

**17-  
AUG-26**

Operator applied downward force directly to the weigh pan while vacuuming. The system was energized.

Repeated direct-load behavior observed.

## DIRECT OBSERVATION

**20-  
AUG-26**

The weigh pan was again held by hand during vacuum cleaning.

Confirms the handling behavior recurred.

## SERVICE HISTORY

**HISTORICAL**

TW#151494 documented a mechanical failure associated with pressing the pan during cleaning.

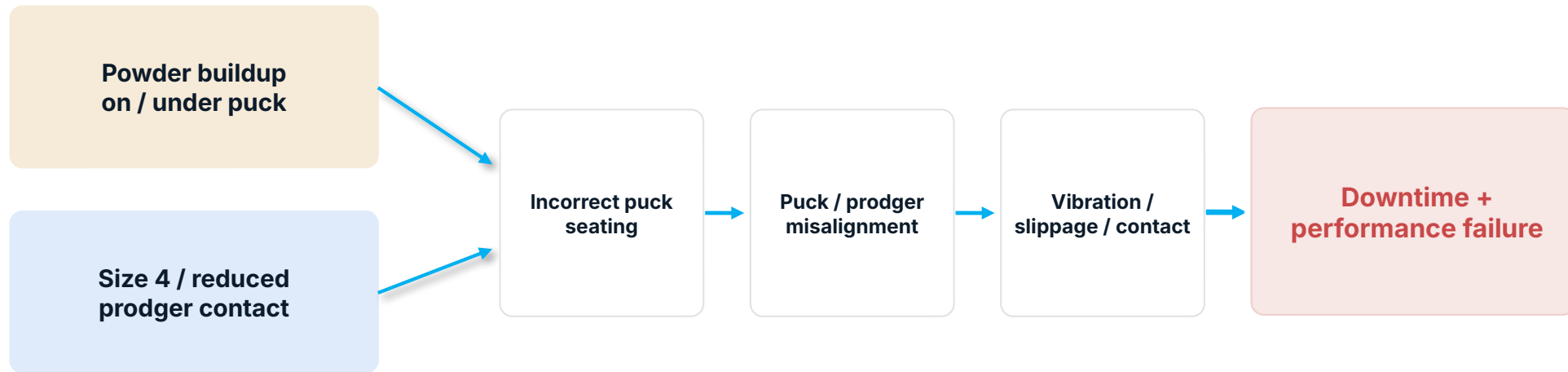
Relevant historical event supports the same load pathway.

What this proves: the loading behavior exists. What still needs manufacturer confirmation: allowable load / damage threshold.

Other:

# Powder and alignment conditions may amplify vibration and cause **Downtime**

This pathway is supported by observed buildup and operator / SME input, but it is not as direct as the pan-loading evidence.



## EVIDENCE

Observed: powder stuck on pucks (17 / 20 AUG) • Reported: Size 4 slippage + incomplete prodger contact  
Observed: pre-alignment performed by 2nd shift • 23-AUG downtime reported as alignment-related

# The same cleaning task can be performed without direct weigh-pan loading

| MANUFACTURING<br>1 <sup>st</sup> / 2 <sup>nd</sup> Shift — 17 / 20 AUG |                                                           | THIRD SHIFT — 31 AUG |                                     | PRODUCT DEVELOPMENT |                                                                |
|------------------------------------------------------------------------|-----------------------------------------------------------|----------------------|-------------------------------------|---------------------|----------------------------------------------------------------|
| WEIGH-PAN CONTACT                                                      | Direct hand restraint / downward force                    | WEIGH-PAN CONTACT    | No hand / vacuum-nozzle contact     | WEIGH-PAN CONTACT   | No abnormal contact observed                                   |
| EQUIPMENT STATE                                                        | Energized during observed cleaning, Weigh Pan not removed | REMOVABLE PARTS      | Removed, wiped, cleaned later       | CLEANING METHOD     | Weigh pan removed before cleaning                              |
| PUCK CONDITION                                                         | Powder buildup observed                                   | ALARMS               | None observed during cleaning       | SME INTERPRETATION  | Operational failure most likely when pan is contacted / loaded |
| RESULTING EVIDENCE                                                     | Credible external load path                               | VALUE OF COMPARISON  | Shows no-contact method is feasible | VALUE OF COMPARISON | Reduces opportunity for direct loading                         |

CONTROLLED LEARNING → Direct pan contact is not required to perform the cleaning activity.

# Three observed gaps matter because each changes the mechanical risk during cle

01

## CONTROL

**Dry wipe + vacuum**

## OBSERVED

**Vacuum-only cleaning, weigh pan not removed observed on 14-AUG**

## RISK LINK

Leaves a greater chance of powder remaining on / around removable parts

02

## CONTROL

**Protect fragile weigh pan from contact**

## OBSERVED

**Pan held / pressed instead of removal during cleaning on 17-AUG and 20-AUG**

## RISK LINK

Creates a direct external mechanical load path

03

## CONTROL

**Clean in required de-energized state**

## OBSERVED

**Observed vacuum cleaning while energized (6.13.1 USTP-608)**

## RISK LINK

Adds unnecessary interaction with the live mechanism during handling

Positive practice to formalize

2nd-shift pre-alignment and no-contact cleaning technique provide practical controls worth standardizing

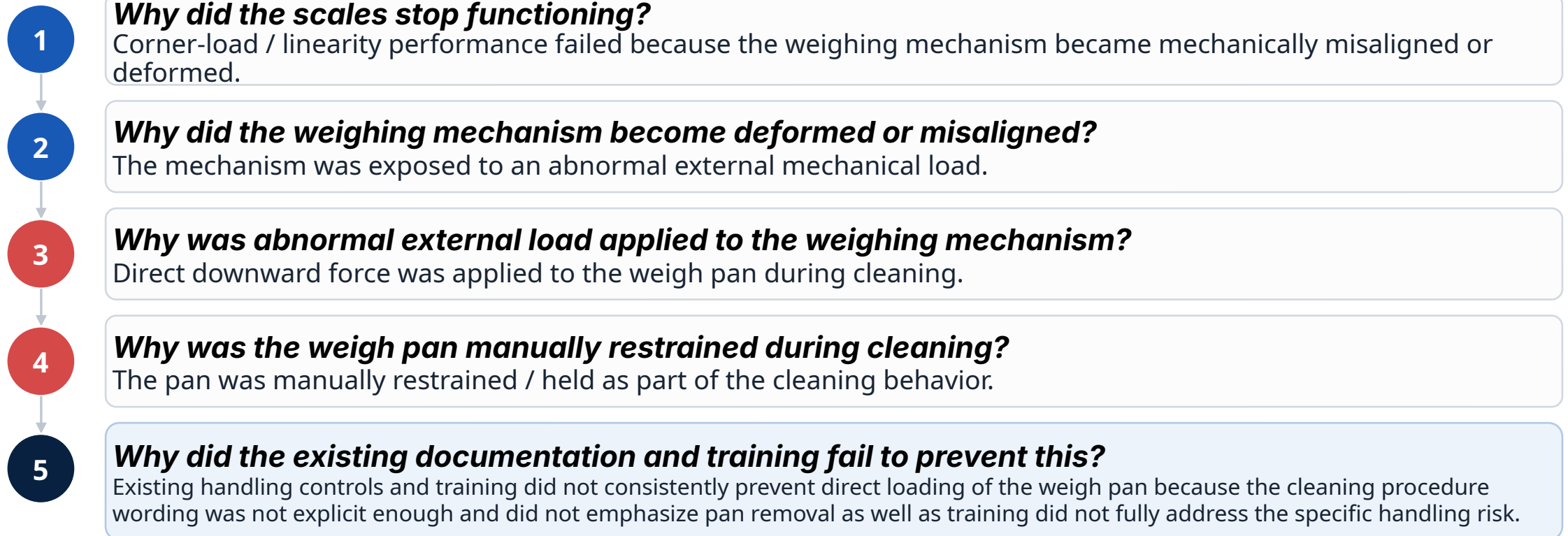
# The hypotheses do not carry equal weight



| <i>HYPOTHESIS</i>                 | <i>STATUS</i>      | <i>WHY</i>                                                     |
|-----------------------------------|--------------------|----------------------------------------------------------------|
| Direct external load on weigh pan | Evidence-supported | Repeated current observations + historical event               |
| Puck / prodger misalignment       | Supported          | Alignment downtime, pre-alignment, SME / operator input        |
| Powder buildup → improper seating | Supported          | Caked / stuck powder observed on pucks                         |
| Operational vibration / contact   | Possible           | Size 4 slippage + scuff evidence; causal link not fully proven |
| Vacuum-related loading / static   | Possible           | Static alarms temporally associated with one cleaning event    |

Use this ranking to focus CAPA on the supported load path while continuing to trend the secondary contributors.

# The 5-Why moves from the failed balance to the control that allowed repeat loading



The system-level failure is not “an operator mistake”; it is an ineffective control that allowed the same fragile-component loading behavior to recur.

# Ineffective control of weigh-pan handling during cleaning

Repeated direct loading of the fragile pan provides the clearest supported path to mechanical deformation / misalignment and corner-load failure.

## Most Probable Root Cause

Direct manual force was repeatedly applied to the weigh pan during cleaning; The existing handling controls and training did not consistently prevent recurrence.

### Mechanical consequence

External load → deformation / misalignment → corner-load and/or linearity failure

## Why this outranks the alternatives

- REPEATED** Direct load observed 17-AUG and 20-AUG
- HISTORICAL** 2023 event links pan pressing with mechanical failure
- PHYSICAL** [redacted] and [redacted] show mechanical abnormalities
- COMPARISON** 3rd shift and PD demonstrate no-contact cleaning is feasible

Secondary contributors remain relevant: puck/prodger alignment, powder buildup, Size 4 slippage/vibration, and vacuum/static behavior should continue to be controlled.

# Observations and Corrective actions

## OBSERVED / DOCUMENTED

- Physical condition of failed balances
- 17 + 20 AUG direct pan loading
- Powder buildup on pucks
- 31 AUG no-contact cleaning
- PD pan-removal practice follows SOP and Weigh Pan removal
- SOP-to-practice differences

## SUPPORTED INFERENCE

- Direct load transfers into weighing mechanism
- Powder can interfere with correct seating
- Misalignment can increase contact / vibration
- No-contact cleaning lowers loading opportunity

## CONCLUSION / CORRECTIVE ACTION

- Most probable root cause: Training outdated and control gap leading to direct pan force
- Leads only should remove and install the microbalance
- Reinforcement and Retraining Initiative
- Pan removal reinforced in SOP, update language. Bold section 6.3.2.11.2 "Remove Weighing pan and spill tray..."; Add Pan removal to section 6.13.2 Minor Cleaning. All for USTP-608
- MSAT Engineers get Method Development/  
Super-User role and Trained Leads given Operator/Super-User

Primary internal sources: Xcelodose investigation, current RCA • Historical findings • 14–31 AUG Manufacturing observations  
• Product Development comparison • SOP review vs Observations

# Control the load path first, then standardize alignment and corrective action



## CONTAIN NOW

### Reduce immediate exposure

- No direct force on weigh pan
- Adjust SOP and required equipment state to include pan removal during cleaning
- Keep pan isolated from vacuum / direct force

## STANDARDIZE

### Remove method variation

- Define cleaning procedure and removable-part cleaning
- Clarify pre-alignment ownership
- Assign MSAT to Method Dev/Super user and remove any Operator from roles with parameter adjustment

## TRAIN

### Make correct handling repeatable

- Demonstrate proper cleaning and correct pan removal
- Qualify operator / super-user alignment role
- Training and SOP updates for Pan removal and no contact cleaning

## ACT

### Close remaining technical gaps

- Update SOP Terminology and Cleaning sections
- Inspect Operations after training
- Installation and Removal of balance by leads only.

Implementation principle → engineer out the contact opportunity where practical; do not rely on memory alone.

# The mechanical pathway is actionable now

Formalize Controls, assign ownership and verify effectiveness

1

**Approve the cleaning / handling control direction**

No direct weigh-pan force; define removal / isolation and equipment-state requirements.

2

**Assign alignment ownership + training**

Decide who can perform pre-alignment, removal and installation;  
As well as how competency is documented.

3

**Implement physical damage-prevention controls**

Lead operator routine inspections. Leads verify retraining effectiveness and document repeat pan contact, misalignment, or damage.

4

**Track recurrence and reliability performance**

Document scale failures, alignment issues, damage events, and repeat maintenance calls and confirm whether the new controls are reducing failures over time.

## CURRENT CONCLUSION

Most probable root cause: ineffective control of direct weigh-pan handling during cleaning.